

**Notes: Angeletos, Lorenzoni & Pavan (RES, 2023)**  
**Wall Street and Silicon Valley: A Delicate Interaction**

## Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                       | <b>3</b>  |
| <b>2</b> | <b>Incremental contribution of the paper</b>                             | <b>3</b>  |
| <b>3</b> | <b>Data and measurement / model primitives</b>                           | <b>4</b>  |
| 3.1      | Technology and fundamental . . . . .                                     | 4         |
| 3.2      | Timing . . . . .                                                         | 4         |
| 3.3      | Information . . . . .                                                    | 4         |
| 3.4      | IPO financing and investor valuation . . . . .                           | 5         |
| 3.5      | Aggregate entry as a signal . . . . .                                    | 5         |
| <b>4</b> | <b>Models and theoretical specifications</b>                             | <b>5</b>  |
| 4.1      | IPO-stage solution . . . . .                                             | 5         |
| 4.2      | Start-up condition . . . . .                                             | 6         |
| 4.3      | Endogenous signal extraction . . . . .                                   | 6         |
| 4.4      | Equilibrium linear rule: Proposition 1 . . . . .                         | 6         |
| 4.5      | Existence, uniqueness, and amplification: Propositions 2 and 3 . . . . . | 6         |
| 4.6      | Mispricing and speculation . . . . .                                     | 7         |
| 4.7      | Beauty contest in real investment: Proposition 4 . . . . .               | 7         |
| 4.8      | Growth potential and finance dependence: Proposition 5 . . . . .         | 7         |
| 4.9      | Planner problem and welfare: Proposition 6 . . . . .                     | 7         |
| 4.10     | Lean-against-the-wind policy: Proposition 7 . . . . .                    | 8         |
| 4.11     | Imperfectly elastic supply of funds: Propositions 8 and 9 . . . . .      | 8         |
| <b>5</b> | <b>Identification and theoretical design</b>                             | <b>8</b>  |
| 5.1      | Mechanism isolated by the baseline model . . . . .                       | 8         |
| 5.2      | Counterfactual without informational spillovers . . . . .                | 9         |
| 5.3      | Counterfactual without IPO-financing sensitivity . . . . .               | 9         |
| 5.4      | Equilibrium versus planner . . . . .                                     | 9         |
| 5.5      | Two sources of inefficiency . . . . .                                    | 9         |
| 5.6      | Robustness and scope . . . . .                                           | 9         |
| <b>6</b> | <b>Figures and result interpretations</b>                                | <b>10</b> |
| 6.1      | Figure 1: IPO waves . . . . .                                            | 10        |
| 6.2      | Figure A1: Improving the start-up rule . . . . .                         | 10        |
| 6.3      | Result box: Strategic complementarity and welfare . . . . .              | 11        |

|          |                                            |           |
|----------|--------------------------------------------|-----------|
| <b>7</b> | <b>Short summary</b>                       | <b>11</b> |
| <b>8</b> | <b>Conclusion</b>                          | <b>12</b> |
| 8.1      | Core conclusion . . . . .                  | 12        |
| 8.2      | Economic intuition . . . . .               | 12        |
| 8.3      | Why the mechanism is inefficient . . . . . | 12        |
| 8.4      | Policy implication . . . . .               | 12        |
| 8.5      | Limits and open questions . . . . .        | 12        |
| 8.6      | Takeaway . . . . .                         | 12        |

# 1 Big picture

- **Question.** Can the interaction between entrepreneurial activity and financial-market beliefs generate excessive waves in start-ups, IPOs, and asset prices even when all agents are rational?
- **Core mechanism.** Silicon Valley and Wall Street form a two-way feedback loop. Entrepreneurs care about IPO prices because those prices determine how much capital they can raise. Investors look at aggregate start-up activity because it contains information about the profitability of new technologies.
- **Main result.** This feedback creates a beauty-contest complementarity in real investment. Entrepreneurs put too much weight on correlated signals, so noise or exuberance generates excessive start-up and IPO waves.
- **Information aggregation.** Aggregate entry  $N$  is informative about fundamentals, but it also reflects correlated noise in entrepreneurs' beliefs. Investors cannot perfectly separate the two.
- **Mispricing in whose eyes?** Investors are rational and do not knowingly overpay. But entrepreneurs may privately expect IPO prices to be high relative to their own information, and this makes entry partly speculative.
- **Welfare.** The equilibrium is inefficient for two linked reasons: a speculative motive in entry and an informational externality in how entry reveals information to guide later investment.
- **Policy.** A policy that “leans against the wind” by taxing or reducing subsidies when IPO prices are high can improve welfare without the government having superior information about fundamentals.
- **Testable prediction.** The mechanism predicts that IPO waves can reflect noise as well as fundamentals. If noise is important, high IPO activity should be associated with lower subsequent IPO returns after conditioning on later information about fundamentals.
- **Economic takeaway.** Financial markets are not merely passive funders of innovation. When real activity itself informs financial markets, prices can feed back into entry decisions and amplify non-fundamental waves.

# 2 Incremental contribution of the paper

- **A new channel between finance and real activity.** The paper combines two forces that are usually studied separately: asset prices affect real investment, and real investment reveals information to asset markets.
- **A rational-expectations theory of IPO waves.** IPO waves need not be driven by irrational investors or exogenous sentiment. They can arise from rational signal extraction when correlated noise in entrepreneurs' information is amplified through the entry-price feedback loop.
- **A real-side beauty contest.** The paper moves the beauty-contest logic from purely financial trading to start-up activity and IPO entry. Entrepreneurs want to know not only what the fundamental is, but also what the market will infer from the activity of other entrepreneurs.
- **A welfare result, not only a positive theory.** The paper shows that the complementarity is inefficient because private entry decisions respond to speculative gains and fail to internalize the information content of aggregate entry.
- **A policy implication without government information advantage.** The recommended policy leans against IPO waves by conditioning on market prices, not by requiring the government to know fundamentals better than private agents.
- **A bridge across literatures.** The paper connects IPO-wave evidence, macro-finance feedback effects, information aggregation, Grossman-Stiglitz style price informativeness, and beauty-contest inefficiencies.

**Interpretation:** The incremental contribution is conceptual. The paper does not simply say that bubbles or IPO waves exist. It explains why rational agents can generate excessive waves when real investment and financial-market beliefs are informationally intertwined.

### 3 Data and measurement / model primitives

This is a theoretical paper. There is no empirical estimation sample. The relevant “data and measurement” objects are model primitives: the fundamental, the entrepreneurs’ signals, the investors’ signals, aggregate entry, IPO prices, and welfare.

#### 3.1 Technology and fundamental

Each entrepreneur who starts a project at  $t = 1$  and raises capital  $k_i$  at  $t = 2$  produces

$$q_i = \Theta f(k_i), \quad f(k) = k^\alpha, \quad \alpha \in (0, 1),$$

where  $\Theta$  is aggregate profitability and  $\theta \equiv \log \Theta$  is normally distributed.

**Interpretation:** The parameter  $\alpha$  measures how much profits depend on the amount of IPO-financed expansion. A high  $\alpha$  makes the entrepreneur more sensitive to financial-market valuations because better IPO prices permit larger capital investment.

#### 3.2 Timing

1. **Nature.** The fundamental  $\theta$  and signals are drawn.
2. **Start-up stage,  $t = 1$ .** Entrepreneur  $i$  chooses  $n_i \in \{0, 1\}$ , where  $n_i = 1$  means starting the project. Aggregate start-up activity is  $N$ .
3. **IPO stage,  $t = 2$ .** The profile of start-up decisions is observed. Active entrepreneurs raise funds by selling shares in IPO markets.
4. **Production and payoff,  $t = 3$ .** The fundamental is revealed and output is realized.

**Interpretation:** The timing generates the two-way feedback. Entrepreneurs move first, but investors later use the aggregate pattern of their decisions to infer profitability and price IPO shares.

#### 3.3 Information

Entrepreneur  $i$  observes two signals:

$$x_i = \theta + \xi_i,$$

with idiosyncratic noise, and

$$y = \theta + \varepsilon,$$

with correlated noise shared by entrepreneurs. Investors observe their own signal, denoted  $\omega$ , and aggregate start-up activity  $N$  at the IPO stage.

**Interpretation:** The correlated noise  $\varepsilon$  is the model’s rational-expectations analogue of market sentiment or exuberance. It is not necessarily irrational: it can come from common networks, common venture capitalists, common industry events, or common interpretation of ambiguous technology news.

### 3.4 IPO financing and investor valuation

At the IPO stage, active entrepreneur  $i$  finances investment by selling shares:

$$k_i = p_i s_i,$$

where  $p_i$  is the share price and  $s_i$  is the share sold. Investors are risk neutral, so in equilibrium

$$p_i = \beta \hat{\Theta} f(k_i),$$

where  $\hat{\Theta} = E[\Theta \mid I]$  is the investors' posterior expectation of profitability.

**Interpretation:** IPO prices are rational from investors' perspective. The potential inefficiency comes from how entrepreneurs anticipate those prices and how their collective entry decisions affect what investors learn.

### 3.5 Aggregate entry as a signal

In equilibrium, the start-up rule is linear:

$$(1 - b)x_i + by \geq c.$$

This implies aggregate entry

$$N = \Phi \left( \frac{\sqrt{\pi_x} \{ (1 - b)\theta + by - c \}}{1 - b} \right),$$

so observing  $N$  is equivalent to observing

$$z = (1 - b)\theta + by = \theta + b\varepsilon.$$

**Interpretation:** Aggregate entry is informative because it aggregates dispersed private signals. But it is noisy because it also aggregates the common noise  $\varepsilon$ . The coefficient  $b$  determines how much sentiment contaminates the signal extracted from entry.

## 4 Models and theoretical specifications

### 4.1 IPO-stage solution

The entrepreneur's relaxed IPO-stage problem implies

$$K(\hat{\Theta}) = \arg \max_k \{ \beta \hat{\Theta} f(k) - k \}.$$

With  $f(k) = k^\alpha$ ,

$$\begin{aligned} K(\hat{\Theta}) &= (\alpha \beta \hat{\Theta})^{\frac{1}{1-\alpha}}, \\ P(\hat{\Theta}) &= \alpha^{\frac{\alpha}{1-\alpha}} (\beta \hat{\Theta})^{\frac{1}{1-\alpha}}, \\ S(\hat{\Theta}) &= \alpha. \end{aligned}$$

**Interpretation:** Better investor beliefs raise IPO prices, which relaxes financing and increases scale. But the amount of shares sold is constant at  $\alpha$ , so IPO-market outcomes do not reveal the entrepreneur's private information beyond what is already revealed by aggregate entry.

## 4.2 Start-up condition

The entrepreneur starts a project if and only if

$$E_i[(1 - \alpha)\theta + \alpha\hat{\theta}] \geq C,$$

where  $\hat{\theta} = E[\theta | I]$  is the investors' posterior mean.

**Interpretation:** The start-up decision depends on both fundamental profitability and market valuation. The term in  $\hat{\theta}$  is the Wall Street channel: high expected IPO valuations encourage entry.

## 4.3 Endogenous signal extraction

Since observing  $N$  is equivalent to observing  $z = \theta + b\varepsilon$ , the precision of the endogenous signal extracted from aggregate entry is

$$\pi_z = \frac{\pi_y}{b^2}.$$

Investors combine  $z$  with their exogenous signal  $\omega$  to form  $\hat{\theta} = E[\theta | z, \omega]$ .

**Interpretation:** A larger  $b$  means entrepreneurs rely more on the correlated signal  $y$ , so entry is more sensitive to common noise. This makes  $N$  less informative about fundamentals and more contaminated by sentiment.

## 4.4 Equilibrium linear rule: Proposition 1

### Proposition 1: Linear equilibrium start-up rule

There exist functions such that if  $b^*$  is a fixed point and  $c^*$  is the corresponding threshold, then there is an equilibrium in which entrepreneur  $i$  enters if and only if

$$(1 - b^*)x_i + b^*y \geq c^*.$$

**Interpretation:** This result reduces the model to two core coefficients:  $b^*$ , the weight on the correlated signal, and  $c^*$ , the entry threshold. Most positive and normative results are about whether equilibrium  $b^*$  is too high.

## 4.5 Existence, uniqueness, and amplification: Propositions 2 and 3

### Proposition 2: IPO waves from rational signal extraction

There is an equilibrium with  $b^* \in (0, 1)$ . In that equilibrium, start-up activity  $N$ , investor valuation  $\hat{\theta}$ , and IPO prices are increasing in both the fundamental  $\theta$  and the common noise  $\varepsilon$ . Equilibrium is unique when  $\alpha$  is small enough; multiple equilibria may arise when  $\alpha$  is large.

### Proposition 3: Information spillovers amplify noise

The informational spillover from real activity to financial markets amplifies the contribution of correlated noise to aggregate volatility.

**Interpretation:** A positive realization of common noise makes entrepreneurs enter more. Investors observe the entry wave and interpret it partly as good news about fundamentals. This raises IPO valuations, which validates and amplifies the entry wave.

## 4.6 Mispricing and speculation

From investors' perspective, prices are conditionally unbiased:

$$E[p - \beta d \mid I] = 0,$$

where  $d = \Theta f(K)$  is the dividend. From entrepreneurs' perspective, however,

$$E_i[p - \beta d]$$

can be positive or negative because entrepreneurs know more about the source of variation in  $N$ .

**Interpretation:** There is no “mistake” by investors in their own information set. Speculation enters because entrepreneurs can forecast when investors are likely to value projects above the entrepreneurs' own assessment of discounted dividends.

## 4.7 Beauty contest in real investment: Proposition 4

### Proposition 4: Coordination-game representation

There exist scalars  $r, c^\# > 0$  such that equilibrium start-up activity can be represented as a binary-action coordination game where entrepreneur  $i$  enters if and only if

$$E_i[(1 - r)\theta + r\Phi^{-1}(N)] \geq c^\#.$$

**Interpretation:** Entrepreneurs care about aggregate entry because aggregate entry affects what investors infer and therefore how IPOs are priced. This is a beauty contest, but the object of coordination is real start-up activity rather than pure financial trading.

## 4.8 Growth potential and finance dependence: Proposition 5

### Proposition 5: High- $\alpha$ sectors are more volatile

A higher  $\alpha$  increases the contribution of correlated noise to aggregate volatility, but only through the information spillover.

**Interpretation:** Sectors with high growth potential or high dependence on IPO financing should be especially prone to non-fundamental waves. This fits the idea that new, uncertain, R&D-intensive sectors are vulnerable to hot IPO markets.

## 4.9 Planner problem and welfare: Proposition 6

A planner choosing a linear entry rule maximizes welfare based on aggregate output net of entry and investment costs. Let  $(b^\diamond, c^\diamond)$  be the welfare-maximizing start-up rule.

### Proposition 6: Equilibrium puts too much weight on common noise

When equilibrium is unique,  $b^\diamond < b^*$ . Therefore, equilibrium entry is too sensitive to the signal with correlated noise, and the contribution of correlated noise to aggregate volatility is inefficiently high.

**Interpretation:** There are two wedges. First, entrepreneurs respond to expected IPO mispricing because selling overvalued shares is privately profitable even though it is a transfer from investors. Second, entrepreneurs do not internalize how their entry rule affects the informativeness of aggregate entry for later investment.

#### 4.10 Lean-against-the-wind policy: Proposition 7

A price-contingent subsidy/tax can implement the socially optimal entry rule:

$$\sigma^*(p) = \frac{1}{\beta} (1 - \exp\{B \ln p - C\})$$

for some  $B > 0$  and  $C \in \mathbb{R}$ .

**Proposition 7: Price-contingent policy can implement the planner's rule**

The socially optimal start-up rule can be implemented by a subsidy/tax paid to entrepreneurs after IPO completion and decreasing in the realized IPO price.

**Interpretation:** A high IPO price tends to reflect either good fundamentals or correlated optimism. The policy does not need to know which is which. By reducing entry incentives when prices are high, it weakens the speculative motive and improves information aggregation.

#### 4.11 Imperfectly elastic supply of funds: Propositions 8 and 9

The baseline assumes a perfectly elastic supply of IPO funds. Section 6 relaxes this by letting investors' opportunity cost be convex. The equilibrium entry condition becomes

$$E_i \left[ \exp \left\{ \frac{1 + \epsilon}{1 - \alpha + \epsilon} \left( (1 - a)\theta + a\hat{\theta} - \psi \ln N \right) \right\} \right] \geq \chi.$$

**Propositions 8 and 9: Robustness to financial crowding out**

Convex funding costs introduce a neoclassical strategic-substitutability term  $-\psi \ln N$ , but the inefficiency still comes only from the speculative motive and informational externality identified in the baseline model.

**Interpretation:** Even if more IPOs make funds more expensive, this does not create the welfare wedge emphasized by the paper. The inefficiency remains the informational/financial feedback between entry and prices.

## 5 Identification and theoretical design

This is a theoretical paper, so “identification” means isolating the mechanism inside the model rather than estimating a causal effect from data.

### 5.1 Mechanism isolated by the baseline model

The baseline model isolates a two-way feedback:

$$\text{entrepreneurial entry} \rightarrow \text{investor beliefs} \rightarrow \text{IPO prices} \rightarrow \text{entrepreneurial entry}.$$

The model removes many other forces: taxes, agency problems, investor irrationality, institutional frictions, and explicit behavioural bubbles.

**Interpretation:** Because agents are rational, any non-fundamental wave is not produced by irrationality. It is produced by rational signal extraction when common noise contaminates aggregate entry.



## 5.2 Counterfactual without informational spillovers

A central benchmark removes the investors' learning from  $N$ . If investors do not infer fundamentals from start-up activity, then entrepreneurs' start-up condition collapses to one based only on their own expectation of  $\theta$ .

$$E_i[\theta] \geq C.$$

**Interpretation:** This benchmark shows that the excessive sensitivity to the common signal does not come merely from entrepreneurs having a common signal. It comes from the fact that Wall Street learns from aggregate entry.

## 5.3 Counterfactual without IPO-financing sensitivity

When  $\alpha$  is small, profits are not very sensitive to IPO-financed expansion. Then the Wall Street valuation channel is weak. As  $\alpha$  rises, entrepreneurs care more about IPO prices, so the feedback loop strengthens.

**Interpretation:** This comparative static identifies the finance-dependence channel. The model predicts stronger waves in sectors where scaling up after IPO is especially important.

## 5.4 Equilibrium versus planner

The planner can choose the start-up rule, while entrepreneurs choose privately. Comparing  $b^*$  to  $b^\diamond$  separates private and social incentives.

$$b^\diamond < b^*.$$

**Interpretation:** The inequality is the model's normative identification result. It says that the market puts too much weight on the common, sentiment-like signal relative to the socially optimal information use.

## 5.5 Two sources of inefficiency

1. **Speculative motive.** Entrepreneurs enter partly because they expect the market to overvalue their projects relative to their own information. This is privately profitable but socially a transfer.
2. **Informational externality.** Entrepreneurs do not internalize how their entry rule affects the informativeness of  $N$  for investors' later capital-allocation decisions.

**Interpretation:** Both wedges originate in the same feedback loop. That is why a single lean-against-the-wind policy can address both.

## 5.6 Robustness and scope

The paper discusses robustness to investor sentiment, secondary markets, partial information revelation at the IPO stage, sequential entry, and imperfectly elastic funding supply.

**Interpretation:** The mechanism requires only two core properties: investors interpret start-up activity as a signal of profitability, and entrepreneurs retain some information advantage about whether entry reflects fundamentals or noise.

## 6 Figures and result interpretations

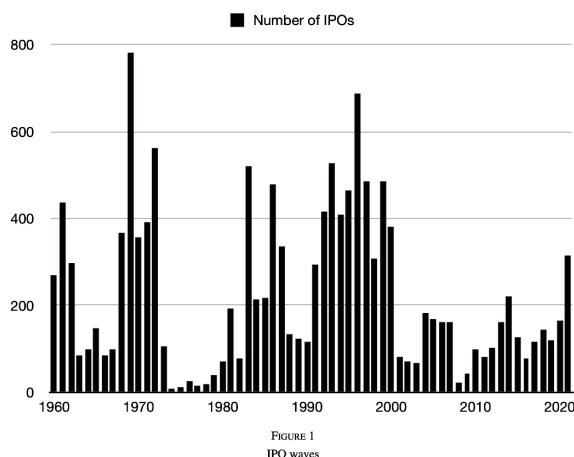
### 6.1 Figure 1: IPO waves

Read:

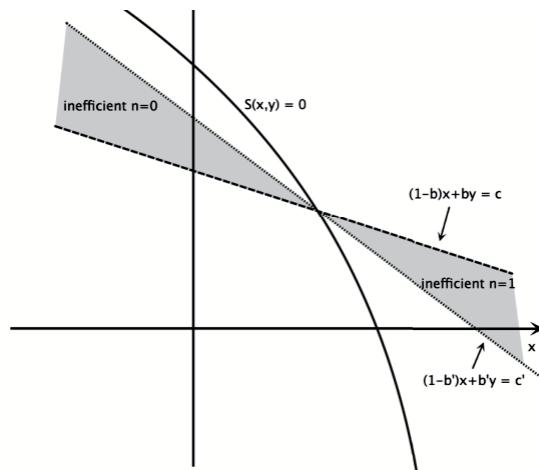
- The bars show large changes in IPO volume across years rather than a smooth trend.
- The spikes coincide with well-known periods of technology or sectoral enthusiasm, especially the dot-com era.
- The post-2000 decline and later revival show that IPO activity can be dormant for long periods and then surge suddenly.

**Economic message:** The model explains how such waves can arise from rational feedback between entry and IPO prices. High activity can reflect fundamentals, but it can also reflect correlated noise that investors partly mistake for fundamentals.

**Interpretation:** The key point is not that every IPO wave is inefficient. The figure motivates a theory in which some waves are informationally excessive because entry itself becomes a signal watched by financial markets. When entry responds to common noise, the signal is contaminated, and prices feed the contamination back into additional entry.



**Figure 1: IPO waves.** The figure plots the number of IPOs per year in the United States from 1960 to 2021. It documents the empirical phenomenon motivating the model: IPOs cluster in waves, with very large spikes around the late 1960s, mid-1980s, mid-to-late 1990s, and 2021. The figure does not prove the model, but it provides the empirical pattern that the theory seeks to rationalize.



**FIGURE A1**  
Improving the start-up rule

**Figure A1: Improving the start-up rule.** The figure illustrates a proof argument behind the planner's preferred start-up rule. The equilibrium line  $(1 - b)x + by = c$  places too much weight on the common signal  $y$ . A modified rule  $(1 - b')x + b'y = c'$  with lower  $b'$  reduces inefficient entry/inaction regions and improves information use.

### 6.2 Figure A1: Improving the start-up rule

Read:

- The axes are the two entrepreneur signals, the idiosyncratic signal  $x$  and the common signal  $y$ .

- The dashed and dotted lines represent different linear entry rules.
- The shaded regions mark choices that are inefficient under the original equilibrium rule.
- Lowering the weight on  $y$  rotates the decision rule toward the idiosyncratic information and reduces sentiment-driven mistakes.

**Economic message:** The figure gives geometric intuition for why the planner prefers  $b^\diamond < b^*$ . The common signal is socially overused because it is correlated across entrepreneurs and therefore makes aggregate entry less informative.

**Interpretation:** The diagram is the appendix proof counterpart of the main welfare result. It shows that the inefficiency is not merely too much or too little entry on average. It is a distortion in the *direction of information use*: private agents rely too much on a signal whose noise is common and therefore creates aggregate waves.

### 6.3 Result box: Strategic complementarity and welfare

#### How the pieces fit together

- Entry uses  $(1 - b)x + by$ .
- Aggregate entry reveals  $z = \theta + b\varepsilon$ .
- Investors raise valuations when  $z$  is high.
- Entrepreneurs anticipate this and increase the equilibrium  $b$ .
- The planner lowers  $b$  to reduce speculation and improve information aggregation.

**Interpretation:** This chain is the paper’s central contribution. The private equilibrium reacts too strongly to the sentiment-like signal because agents correctly forecast that the financial market will interpret entry as news.

## 7 Short summary

- The paper studies a two-way feedback between start-up activity and asset-market beliefs.
- Entrepreneurs decide whether to start projects using idiosyncratic and common private signals.
- Investors observe aggregate entry and use it as a signal about the profitability of the new technology.
- High entry raises investor beliefs and IPO valuations, which raises the incentive to enter.
- This creates a beauty-contest complementarity in real investment.
- Correlated noise in entrepreneurs’ information can therefore generate excessive waves in start-up activity, IPOs, prices, and investment.
- Investors are rational; mispricing exists only relative to entrepreneurs’ private information.
- The equilibrium puts too much weight on the common, sentiment-like signal.
- The planner prefers a start-up rule with lower sensitivity to the common signal.
- A price-contingent policy that leans against high IPO prices can implement a better rule.
- The policy does not require the planner to know fundamentals better than the market.
- The mechanism is robust to imperfectly elastic funding supply, secondary markets, and partial information revelation.

## 8 Conclusion

### 8.1 Core conclusion

The paper shows that the interaction between Silicon Valley and Wall Street can generate excessive waves in entry, IPOs, and asset prices. Entrepreneurs care about IPO prices because these prices determine how much capital they can raise and how much of future output they retain. Investors, in turn, care about start-up activity because it is an informative signal about profitability. This feedback makes common noise in entrepreneurs' information socially costly.

### 8.2 Economic intuition

The intuitive mechanism is simple but powerful. Suppose many entrepreneurs receive a common optimistic signal about a new technology. Even if that optimism is partly noise, entry rises. Investors see the entry wave and rationally infer that profitability may be high. IPO prices rise. Entrepreneurs anticipated exactly this inference, so the initial optimistic signal becomes even more influential in their entry decisions. The economy can therefore experience an IPO wave that is rational at every private decision node but excessive from a social perspective.

### 8.3 Why the mechanism is inefficient

The inefficiency is not that investors are fooled in a naive sense. Investors price assets rationally given their information. The problem is that entrepreneurs privately gain from issuing equity when the market valuation is high relative to their own information. This speculative motive is a transfer, not a social benefit. In addition, entrepreneurs do not internalize that putting more weight on common noise makes aggregate entry a worse signal for investors. These two wedges make equilibrium entry too sensitive to correlated noise.

### 8.4 Policy implication

The paper's policy result is subtle. The government does not need to know whether an IPO boom reflects true fundamentals or sentiment. A price-contingent tax/subsidy that leans against high IPO prices can reduce the private return to speculative entry and improve the information content of entry. This differs from standard stabilization policy: the target is not aggregate demand or asset-price levels per se, but the information externality created by the price-entry feedback loop.

### 8.5 Limits and open questions

The model is stylized. It abstracts from agency frictions inside start-ups, rich VC contracting, heterogeneous investors, learning over time, and institutional features of IPO markets. It also does not estimate the size of the welfare loss in data. Its contribution is to clarify a mechanism and derive sharp implications. Empirical work could test the prediction that high IPO activity is negatively related to subsequent realized IPO returns after controlling for fundamentals, and could measure whether high-growth sectors with stronger financing needs show greater sensitivity to sentiment-like signals.

### 8.6 Takeaway

The paper reframes IPO waves as a rational-expectations information problem. Hot IPO markets can be informative, but they can also be overly informative in the wrong direction: too much common noise is amplified through the feedback from entry to prices and back to entry. This is why Wall Street and Silicon Valley interact delicately, and why leaning against the wind can raise welfare even without superior government information.